POTASSIUM CITRATE

Hyperkalaemia in renal failure

Dilute — GI irritation

CHEMICAL NAME Tripotassium 2-hydroxypropane-1,2,3-tricarboxylate ($K_3C_6H_5O_7 \cdot H_2O$)
CLASS Urinary alkaliniser; potassium salt
TRADE NAMES Urocit-K · Effercitrate · Uralyt-U · Potassium citrate mixture

THERAPEUTIC USES / INDICATIONS

- Prevention of kidney stones: calcium oxalate (hypocitraturia), uric acid and cystine stones.
- Relief of cystitis symptoms (urinary alkalinisation — burning).
- Renal tubular acidosis with stones; hypokalaemia with metabolic acidosis.

ADMINISTRATION

- **PO** effervescent tablets/granules or mixture well diluted in water, with or after meals.
- ER tablets swallowed whole with food.
- Monitor urine pH (target 6.0–7.0).

DOSAGES

- **Stone prevention:** 30–60 mEq/day in 2–3 divided doses (e.g. 10–20 mEq three times daily).
- **Cystitis relief (mixture):** 10 mL diluted three times daily for up to 48 h.
- **Uric acid stones:** titrate to urine pH 6.5–7.

PHARMACOKINETICS

- **Onset:** urine alkalinisation within hours.
- Citrate metabolised to bicarbonate.
- Excreted in urine (K^+, citrate).

PHARMACODYNAMICS

- Citrate → bicarbonate → alkalinises urine and ↑ urinary citrate (binds calcium, inhibits crystal formation); ↑ uric acid and cystine solubility.
- Provides potassium.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- GI upset: nausea, diarrhoea, abdominal discomfort.
- Mild diuresis.

ADVERSE EFFECTS (SERIOUS)

- Hyperkalaemia (renal impairment, K^+-sparing drugs, ACEi/ARBs).
- GI ulceration (solid ER tablets in delayed gastric emptying).
- Metabolic alkalosis.

CONTRAINDICATIONS & CAUTIONS

- Hyperkalaemia or conditions predisposing (renal failure, Addison's, severe dehydration).
- Active UTI with urease-producing organisms (struvite stones).
- Delayed gastric emptying, oesophageal compression (solid forms).

KEY INTERACTIONS

- **K^+-sparing diuretics, ACE inhibitors, ARBs, NSAIDs, trimethoprim:** hyperkalaemia.
- **Anticholinergics:** ↑ GI injury (solid forms).
- **Methenamine:** inactivated in alkaline urine.

NURSING MANAGEMENT

- Check renal function and K^+ before and during treatment.
- Monitor urine pH and stone recurrence.
- Encourage fluid intake (2.5–3 L/day).
- Cystitis: send urine culture if symptoms persist.

PATIENT EDUCATION

- Dissolve well in water and take after meals.
- Drink plenty of water daily.
- Report palpitations, weakness, tingling (high potassium).
- See a doctor if cystitis symptoms last >48 h or with fever.

KEY POINTS

- Urinary alkaliniser; ↑ urinary citrate.
- Relieves cystitis burning.
- Prevents calcium, uric acid, cystine stones.
- Hyperkalaemia risk; check K^+ and renal function.

SOURCES CHECKED [713] Potassium Citrate Mixture (UK SmPC) [714] Potassium Citrate, oral (US label) [715] Potassium Citrate, oral (US label, Actavis Pharma)
Full references p. 432–457